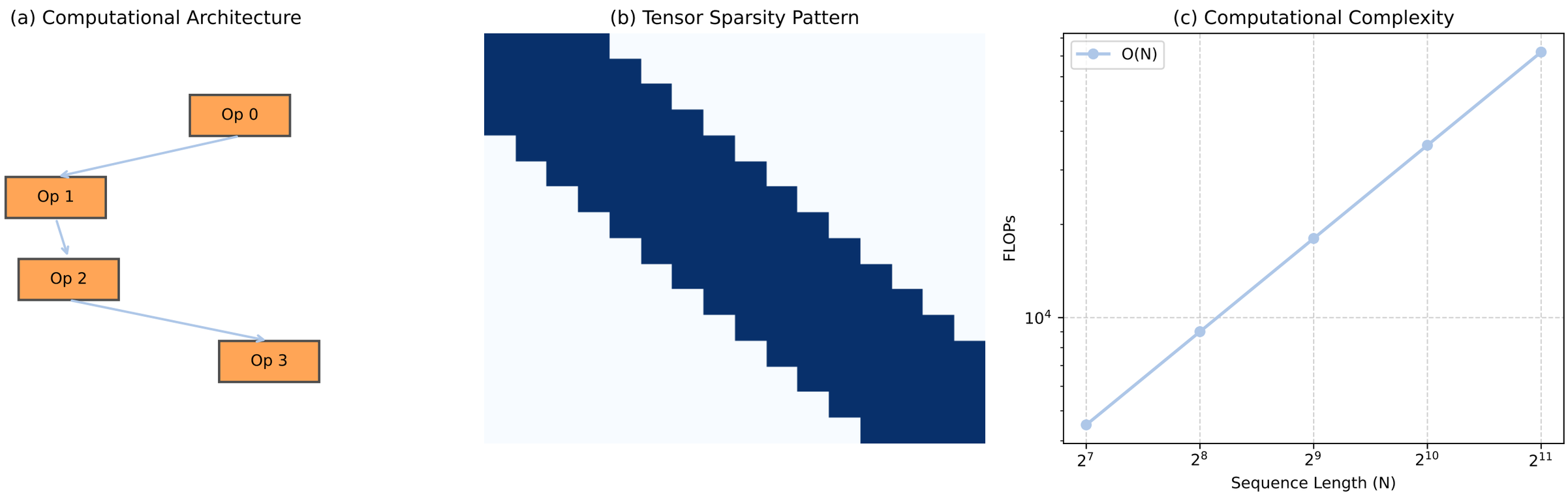Module Architecture: quality_based